

Introduction

The collected dataset contains approximately 2,000 Goodreads book records, including information such as book name, author, meta score, number of votes, serial number, average rating, and rating count. After data cleaning, textual values such as ratings, votes, and scores can be converted into numerical features suitable for machine-learning and statistical analysis. The following four proposals use the cleaned dataset for different analytical and predictive purposes.

Proposal 1: Book Rating Prediction

Objective

The first proposal is to develop a model that predicts the average rating of a book based on other available information about that book.

Dataset Utilization

The following cleaned attributes can be used:

Input features:

- Meta Score
- Number of Voted
- Rating Count
- Author
- Serial Number

Target variable:

- Average Rating

For example, information about a book's score, number of votes, and popularity can be used to estimate its average Goodreads rating.

Model to Be Implemented

Multiple Linear Regression. The model will learn the relationship between the independent variables and the average rating.

The general concept is:

- Meta Score
- Number of Votes
- Rating Count
- Serial Number
- Regression Model
- Predicted Average Rating

Expected Output

The model will produce:

- Predicted average rating for each book
- Actual vs. predicted rating comparison
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 score
- Regression/actual-vs-predicted graph

Expected Result

The model should show whether Goodreads popularity and scoring information can be used to predict a book's average rating.

Proposal 2: Book Popularity Prediction

Objective

The second proposal is to determine which books are likely to become **highly popular** based on their Goodreads information.

Instead of predicting an exact number, books can be classified into categories such as:

- High Popularity
- Low Popularity

Dataset Utilization

The following features can be used:

Input features:

- Meta Score
- Average Rating
- Serial Number
- Number of Votes

Target:

- Popularity category based on Rating Count

For example, the books can be divided using a threshold or percentile:

- High Rating Count → Popular
- Low Rating Count → Less Popular

Model to Be Implemented

Random Forest Classification. Random Forest is suitable because it can handle multiple numerical features and determine which features contribute most to the classification.

The process will be:

- Book Features
- Random Forest Classifier
- High Popularity / Low Popularity

Expected Output

The model will produce:

- Popularity prediction
- Classification accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Feature importance graph

Expected Result

The analysis should identify the characteristics associated with highly popular books and determine which features are most useful for predicting popularity.

Proposal 3: Book Clustering and Recommendation Groups

Objective

The third proposal is to group similar books based on their ratings, scores, and popularity.

Instead of predicting a specific value, this proposal will discover **natural groups of similar books** within the dataset.

For example, the model might produce groups such as:

- Cluster 1 → Highly rated & highly popular books
- Cluster 2 → Highly rated but less popular books
- Cluster 3 → Moderately rated books
- Cluster 4 → Lower-rated / less popular books

These groups can then be used as a basic foundation for a **book recommendation system**.

Dataset Utilization

The following numerical features can be used:

- Average Rating
- Rating Count
- Meta Score
- Number of Votes

The features should be standardized before clustering because they have very different numerical scales.

Model to Be Implemented

K-Means Clustering

The process will be:

- Cleaned Goodreads Data
- Feature Selection
- Data Standardization
- K-Means Clustering
- Book Groups

Expected Output

The model will produce:

- Cluster number for every book
- Similarity-based book groups

- Cluster centers
- Cluster visualization
- List of books belonging to each cluster

Expected Result

The analysis will reveal different types of books within the Goodreads dataset. Books in the same cluster will have similar rating, popularity, and scoring characteristics.

This can later be extended into a simple recommendation system—for example, recommending books from the same cluster as a user's preferred book.

Proposal 4: Author and Book Success Analysis

Objective

The fourth proposal is to investigate which authors and book characteristics are associated with successful books. Unlike the first three proposals, this analysis focuses more on discovering relationships and important patterns in the dataset.

Dataset Utilization

The analysis will use:

- Author
- Book Name
- Average Rating
- Rating Count
- Meta Score
- Number of Votes

The data can be aggregated by author to calculate:

- Number of books per author
- Average rating per author
- Total rating count
- Average Meta Score
- Average number of votes

Model to Be Implemented

Random Forest Regression + Feature Importance Analysis. A Random Forest Regression model can be used to investigate which numerical features are most strongly associated with book popularity/rating. In addition, author-level descriptive statistics will be used to identify high-performing authors.

Expected Output

The analysis will produce:

- Top authors by average rating
- Top authors by number of books
- Top authors by popularity
- Average rating by author
- Relationship between rating and popularity
- Feature importance ranking
- Scatter plots and bar charts

For example:

Feature Importance

- Rating Count
- Meta Score
- Number of Votes
- Average Rating

The actual importance values will be calculated from the dataset.

Expected Result

The analysis should identify the factors most strongly associated with book success and highlight authors whose books perform particularly well in terms of rating and popularity.